

Laboratory Techniques & Measurements

Measurements are important.

- it is how we quantify the world around us.

Need to make measurements of accuracy & precision

- Digital vs analog measurements

↓
gives you all
the significant
digits for measurement

↘ requires operator skill

- knowledge of the scale
- read the divisions on scale
- estimate one digit past the smallest marking
- uncertain digit

Exploration

- Pay special attention to analog measurements under significant figures & error section

- look at the ruler picture especially.

- most common analog measurement is volume

- pay attention to meniscus

Experimentation

- length - make sure to estimate one digit past the smallest marking (and make sure to do the measuring in metric)

Exercise 2

- Record the mass of the graduated cylinder empty
- fill to 5.0 mL w/ H_2O , then weigh full
- the difference between the two measurements of mass is the mass of water
 - should be around 5g
- Repeat with the isopropyl alcohol
- Measure volume by direct measurement in cm^3 should be similar to its volume in mL by displacement

Exercise 3

- 8.0g is just what you want to get if you go over or under slightly that's ok.
 - we generally mean ± 1 of last digit
 - so 7.9g - 8.1g is ok.

det density of this soln

But you must be exact w/ the water!

- you then dilute the 1st or "stock" soln (soln 0) by 2.5 mL of stock in 25 mL of soln

det the density of this soln

- use $C_1 V_1 = C_2 V_2$ to det concentration of the new soln

$$\begin{array}{ll} C_1 = \text{concentration of stock} & V_2 = 25 \text{ mL} \\ V_1 = 2.5 \text{ mL} & C_2 = ? \end{array}$$

- Continue w/ the other soln det the concentration,
by $C_1 V_1 = C_2 V_2$

eg. Soln 0 is stock $8.0g/25 mL = C_1$

Soln 1 is $C_1 V_1 = C_2 V_2$

Det.
density of
each soln,

2.5 mL

25 mL

what you
are calculating

Plot density
and

Soln 2 is

$C_1 V_1 = C_2 V_2$

25 mL

Concentration

4.5 mL

what you are calculating

Soln 3 $V_1 = 3.0 mL$

Soln 4 $V_1 = 6.0 mL$

Remember the rules
of graphing

Questions : $C_1 V_1 = C_2 V_2$ can be used for 1